

B6 Cheatsheet Notes

Readable LaTeX transcription of the handwritten cheatsheet

This note is a readability transcription of the handwritten cheatsheet. The derivation order and formula content have been preserved; headings and spelling are only cleaned up where this helps reading.

1 Crystal Definitions

Lattice	An infinite set of points defined by integer sums of linearly independent primitive cells.
Unit cell	Any region of space that can be repeated over and over again to reconstruct the lattice.
Primitive unit cell	Unit cell with only one atom point.
Wigner–Seitz cell	A space that is closer to one lattice point than to any other lattice point.
Basis	Description of an object in a unit cell with respect to a reference point.
Coordination number	Number of nearest neighbours any point of the lattice has.
Orthorhombic lattice	Orthogonal vectors, all of different length.
Tetragonal lattice	Orthogonal vectors, two of the same length.
Reciprocal lattice	Points denoted by $\mathbf{G}$ satisfying $e^{i\mathbf{G}\cdot\mathbf{R}} = 1 \quad \text{for all } \mathbf{R} \text{ on the direct lattice.}$
Lattice plane	Plane containing at least three non-collinear lattice points.
Family of lattice planes	An infinite set of equally separated parallel planes that contains all points of the lattice.
Brillouin zone	Unit cell in k -space.

2 Cubic Lattices

Here z is the coordination number and A is the number of atoms in the cell.

Lattice	Coordination number z	Atoms in cell A
SC	$z = 6$	$A = 1$
BCC	$z = 8$	$A = 2$
FCC	$z = 12$	$A = 4$

Basis points:

$$\begin{aligned} \text{BCC: } & (0, 0, 0), \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right), \\ \text{FCC: } & (0, 0, 0), \left(0, \frac{1}{2}, \frac{1}{2}\right), \left(\frac{1}{2}, 0, \frac{1}{2}\right), \left(\frac{1}{2}, \frac{1}{2}, 0\right). \end{aligned}$$

3 Reciprocal Lattice and Miller Planes

$$e^{i\mathbf{G}\cdot\mathbf{R}} = 1, \quad \mathbf{G} = h\mathbf{b}_1 + k\mathbf{b}_2 + l\mathbf{b}_3, \quad \mathbf{R} = n_1\mathbf{a}_1 + n_2\mathbf{a}_2 + n_3\mathbf{a}_3.$$

$$\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\delta_{ij}.$$

In three dimensions,

$$\mathbf{b}_i = \frac{2\pi(\mathbf{a}_j \times \mathbf{a}_k)}{\mathbf{a}_i \cdot (\mathbf{a}_j \times \mathbf{a}_k)}.$$

In two dimensions, replace $\mathbf{a}_k$ by $\mathbf{e}_z$.

For orthogonal axes,

$$|\mathbf{b}_i| = \frac{2\pi}{|\mathbf{a}_i|}.$$

For an (hkl) plane, the intercepts are

$$\frac{\mathbf{a}_1}{h} : \frac{\mathbf{a}_2}{k} : \frac{\mathbf{a}_3}{l}.$$

With

$$\mathbf{r}_1 = \frac{\mathbf{a}_1}{h} - \frac{\mathbf{a}_2}{k}, \quad \mathbf{r}_2 = \frac{\mathbf{a}_2}{k} - \frac{\mathbf{a}_3}{l},$$

one has

$$\mathbf{G} \cdot \mathbf{r}_1 = \mathbf{G} \cdot \mathbf{r}_2 = 0,$$

so $\mathbf{G}$ is perpendicular to the plane.

For adjacent planes,

$$\mathbf{G} \cdot \mathbf{r} = 2\pi m, \quad \mathbf{G} \cdot (\mathbf{r}_1 - \mathbf{r}_2) = 2\pi.$$

Therefore

$$d = \frac{2\pi}{|\mathbf{G}|}.$$

Using $\mathbf{G} = h\mathbf{b}_1 + k\mathbf{b}_2 + l\mathbf{b}_3$,

$$d = \frac{2\pi}{|\mathbf{G}|} = \frac{2\pi}{\sqrt{h^2|\mathbf{b}_1|^2 + k^2|\mathbf{b}_2|^2 + l^2|\mathbf{b}_3|^2}}.$$

For a cubic lattice,

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}.$$

4 Diffraction

$$\Gamma = \frac{2\pi}{\hbar} \delta(E_{k'} - E_k) |\langle k'|V|k \rangle|^2.$$

$$\begin{aligned} \langle k'|V|k \rangle &= \frac{1}{L^3} \int d^3r V(\mathbf{r}) e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}} \\ &= \frac{1}{L^3} \sum_{\mathbf{R}} \int_{\text{unit cell}} d^3r V(\mathbf{r}) e^{-i\Delta\mathbf{k}\cdot\mathbf{r}}. \end{aligned}$$

With $V(\mathbf{r} + \mathbf{R}) = V(\mathbf{r})$ and $\mathbf{r} = \mathbf{r}' + \mathbf{R}$,

$$\langle k'|V|k \rangle = \frac{1}{L^3} \sum_{\mathbf{R}} e^{-i\Delta\mathbf{k}\cdot\mathbf{R}} \int_{\text{cell}} d^3r' V(\mathbf{r}') e^{-i\Delta\mathbf{k}\cdot\mathbf{r}'}$$

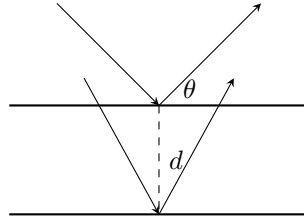
If $e^{-i\Delta\mathbf{k}\cdot\mathbf{R}} \neq 1$, phases add up to zero. Thus

$$e^{i\Delta\mathbf{k}\cdot\mathbf{R}} = 1, \quad \boxed{\Delta\mathbf{k} = \mathbf{G}}.$$

4.1 Bragg and Laue

Extra distance:

$$2d \sin \theta = n\lambda.$$



Equivalence with Laue:

$$\begin{aligned}\hat{e}_k \cdot \hat{G} &= -\hat{e}_{k'} \cdot \hat{G} = \sin \theta, \\ \mathbf{G} = \mathbf{k} - \mathbf{k}' &= \frac{2\pi}{\lambda} (\hat{e}_k - \hat{e}_{k'}).\end{aligned}$$

Therefore

$$|\mathbf{G}| = \frac{4\pi \sin \theta}{\lambda} \implies \frac{2\pi}{|\mathbf{G}|} (2 \sin \theta) = \lambda,$$

and hence

$$\lambda = 2d \sin \theta.$$

5 Structure Factor and Selection Rules

$$S_{hkl}(\mathbf{G}) = \int_{\text{cell}} d^3x e^{-i\mathbf{G} \cdot \mathbf{x}} V(\mathbf{x}).$$

For X-rays,

$$V(\mathbf{x}) = \sum_j Z_j g_j(\mathbf{x} - \mathbf{x}_j).$$

Hence

$$S_{hkl}(\mathbf{G}) \sim \sum_j f_j(\mathbf{G}) e^{i\mathbf{G} \cdot \mathbf{x}_j},$$

or, using fractional coordinates,

$$S_{hkl} = \sum_j f_j e^{i2\pi(hx_j + ky_j + lz_j)}.$$

The sum over j is over different atom types and locations.

$$S_{hkl} = S^{\text{lattice}} S^{\text{basis}},$$

where the lattice factor gives the selection rule and the basis factor gives the different atom type contribution.

$$S_{hkl}^{\text{SC}} = \sum_j f_j \quad \text{for all reflections.}$$

$$S_{hkl}^{\text{BCC}} = \sum_j f_j [1 + (-1)^{h+k+l}],$$

allowed when $h + k + l$ is even.

$$S_{hkl}^{\text{FCC}} = \sum_j f_j [1 + (-1)^{h+k} + (-1)^{k+l} + (-1)^{h+l}],$$

allowed when h, k, l are all odd or all even.

Intensity:

$$I_{hkl} = M_{hkl} |S_{hkl}|^2.$$

Practical X-ray notes:

measured angle is 2θ ,

$$d = \frac{\lambda}{2 \sin \theta} = \frac{a}{\sqrt{N}}, \quad N = h^2 + k^2 + l^2,$$

$$N^2 = \left(\frac{2a \sin \theta}{\lambda} \right)^2.$$

Density/lattice parameter relation:

$$n = \frac{\rho}{m} = \frac{\rho}{A m_B}, \quad n a_0^3 = \text{atoms in cell}.$$

6 Heat Capacity

Models:

1. Sommerfeld: $C \propto T/T_F$.
2. Einstein: $C \propto e^{-\beta \hbar \omega}$.
3. Debye.

6.1 Sommerfeld

$$\varepsilon_F = \frac{\hbar^2 k_F^2}{2m}, \quad \bar{n}_i = \begin{cases} 1, & k < k_F, \\ 0, & k > k_F. \end{cases}$$

With the factor of 2 for spin,

$$N = 2 \frac{V}{(2\pi)^3} \int d^3 k \bar{n}_i = \frac{V}{4\pi^3} \left(\frac{4}{3} \pi k_F^3 \right).$$

Therefore

$$k_F = (3\pi^2 n)^{1/3}, \quad \varepsilon_F = \frac{\hbar^2}{2m} (3\pi^2 n)^{2/3}.$$

Since

$$g(\varepsilon) \propto \varepsilon^{1/2}, \quad N = \int_0^{\varepsilon_F} g(\varepsilon) d\varepsilon \propto \varepsilon_F^{3/2} = C \varepsilon_F^{3/2},$$

$$g(\varepsilon_F) = \frac{dN}{d\varepsilon_F} = \frac{3C \varepsilon_F^{1/2}}{2} = \frac{3C \varepsilon_F^{3/2}}{2\varepsilon_F} = \frac{3N}{2\varepsilon_F}.$$

With $\Delta E \sim k_B T$,

$$\Delta N_{\text{excite}} \sim g(\varepsilon_F) \Delta E \sim g(\varepsilon_F) k_B T,$$

$$\Delta U = \Delta E \Delta N = (k_B T)^2 g(\varepsilon_F),$$

$$C_V = \frac{3N k_B^2 T}{\varepsilon_F} \propto T.$$

6.2 Debye

Assumptions: simple harmonic motion, dispersion $\omega = vk$ up to ω_d , and three modes for each k .

$$g(k) dk = 3 \frac{V}{(2\pi)^3} 4\pi k^2 dk.$$

Using $\omega = vk$,

$$g(\omega) d\omega = \frac{3V}{2\pi^2 v^3} \omega^2 d\omega.$$

The total number of degrees of freedom is

$$3N = \int_0^{\omega_d} g(\omega) d\omega = \frac{3V}{2\pi^2 v^3} \frac{1}{3} \omega_d^3.$$

Thus

$$\frac{3V}{2\pi^2 v^3} = \frac{9N}{\omega_d^3} = \frac{9N\hbar^3}{k_B^3 T_D^3}, \quad \omega_d = \frac{k_B T_D}{\hbar}.$$

$$g(\omega) = \frac{9N\hbar^3}{k_B^3 T_D^3} \omega^2.$$

Mean energy:

$$\begin{aligned} \langle E \rangle &= \int_0^{\omega_d} d\omega g(\omega) \hbar \omega \left(\frac{1}{e^{\beta \hbar \omega} - 1} + \frac{1}{2} \right) \\ &= \frac{9N}{\omega_d^3} \int_0^{\omega_d} d\omega \frac{\hbar \omega^3}{e^{\beta \hbar \omega} - 1} + T \text{ independent} \\ &= \frac{9N\hbar}{\omega_d^3} \frac{1}{(\beta \hbar)^4} \int_0^{T_D/T} dx \frac{x^3}{e^x - 1} + (\dots) \\ &= \frac{9Nk_B T^4}{T_D^3} \int_0^{T_D/T} dx \frac{x^3}{e^x - 1} + (\dots). \end{aligned}$$

If $T_D/T \gg 1$,

$$\langle E \rangle = \frac{3N\pi^4 k_B T^4}{5T_D^3}, \quad C_V = \frac{12\pi^4}{5} Nk_B \left(\frac{T}{T_D} \right)^3.$$

If $T_D/T \ll 1$,

$$\frac{x^3}{e^x - 1} \sim x^2, \quad C_V = 3Nk_B.$$

7 Magnetism

7.1 Paramagnetism

$$E = -\boldsymbol{\mu} \cdot \mathbf{B}, \quad \boldsymbol{\mu} = -g\mu_B \mathbf{S}, \quad M = \frac{1}{2}\mu_B.$$

$$E = g\mu_B B S = \pm \frac{1}{2} g\mu_B B = \pm \mu_B B.$$

$$Z = \sum_j e^{-\beta E_j}, \quad F = -k_B T \ln Z,$$

$$\langle \mu \rangle = \frac{\sum_i \mu_i e^{-\beta E_i}}{Z} = \frac{\mu_B (e^{\beta \mu_B B} - e^{-\beta \mu_B B})}{2 \cosh(\beta \mu_B B)} = \mu_B \tanh \left(\frac{\mu_B B}{k_B T} \right).$$

$$M = n\mu = \frac{N\mu}{V}.$$

For $\beta \mu_B B \ll 1$,

$$\chi_{\text{Curie}} = \mu_0 \frac{dM}{dB} = \frac{n\mu_0 \mu_B^2}{k_B T}.$$

7.2 Larmor Diamagnetism

$$\delta H = \frac{e^2}{8m_e} |\mathbf{B} \times \mathbf{r}|^2.$$

$$\delta E = \frac{e^2 B^2}{8m_e} \langle x^2 + y^2 \rangle = \frac{e^2 B^2 r^2}{12m_e}.$$

$$\mu = -\frac{\partial \delta E}{\partial B} = -\frac{e^2 r^2 B}{6m_e}.$$

$$M = n\mu = Zn\mu \quad \text{for } Z \text{ electrons,}$$

$$\chi = -\frac{Zne^2 r^2 \mu_0}{6m_e}.$$

7.3 Spontaneous Magnetism

$$H = -\frac{1}{2} \sum_{\langle i,j \rangle} J \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_j + g\mu_B \mathbf{B} \cdot \sum_i \boldsymbol{\sigma}_i.$$

Look at site i :

$$H_i = \left(g\mu_B \mathbf{B} - J \sum_j \boldsymbol{\sigma}_j \right) \cdot \boldsymbol{\sigma}_i.$$

Define

$$H_i = g\mu_B B_{\text{eff}} \sigma_i, \quad B_{\text{eff}} = B - \frac{J}{g\mu_B} \sum_j \sigma_j.$$

Then

$$\begin{aligned} Z &= e^{-\beta\mu_B B_{\text{eff}}} + e^{\beta\mu_B B_{\text{eff}}}, \\ \langle \sigma_i \rangle &= \frac{\sum \sigma_i e^{-\beta E_i}}{Z_i} \\ &= \frac{\frac{1}{2} e^{-\beta\mu_B B_{\text{eff}}} - \frac{1}{2} e^{\beta\mu_B B_{\text{eff}}}}{2 \cosh(\beta\mu_B B_{\text{eff}})} \\ &= -\frac{1}{2} \tanh(\beta\mu_B B_{\text{eff}}). \end{aligned}$$

Let $\sigma_j = \sigma_i$. Then

$$\langle \sigma_i \rangle = -\frac{1}{2} \tanh \left(\beta\mu_B B - \frac{1}{2} \beta J z \langle \sigma_i \rangle \right),$$

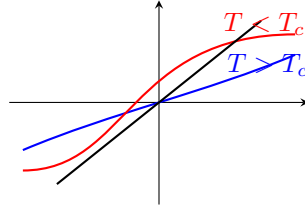
where z is the coordination number.

Self-consistent case $B = 0$:

$$\langle \sigma_i \rangle = -\frac{1}{2} \tanh \left(\frac{1}{2} \beta J z \langle \sigma_i \rangle \right).$$

At T_c , use $\tanh x \sim x$ at $x = 0$:

$$1 = \frac{1}{2} \beta J z = \frac{Jz}{2k_B T_c}, \quad T_c = \frac{1}{4} \frac{Jz}{k_B}.$$



7.4 Curie–Weiss

For $\beta\mu_B B \ll 1$: small B , high T , paramagnet.

$$\begin{aligned} \tanh x &\sim x. \\ \sigma &= -\frac{1}{2} \left(\beta\mu_B B - \frac{1}{2} \beta J z \sigma \right). \end{aligned}$$

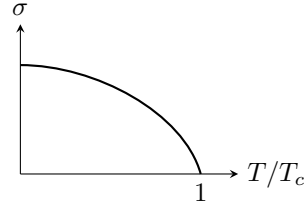
Therefore

$$\sigma = \frac{-\frac{1}{2} \beta\mu_B B}{1 - \frac{1}{4} \beta J z} = \frac{-\frac{1}{2} \mu_B B}{k_B \left(T - \frac{Jz}{4k_B} \right)} = \frac{-\frac{1}{4} g\mu_B B}{k_B (T - T_c)}.$$

$$m = -g\mu_B \langle \sigma \rangle, \quad M = -ng\mu_B \langle \sigma \rangle.$$

$$\chi_{\text{Curie-Weiss}} = \mu_0 \frac{dM}{dB} = \frac{\mu_0 \mu_B^2}{k_B T} \frac{1}{1 - T_c/T}.$$

For a ferromagnet:



7.5 Pauli Paramagnetism Near the Fermi Surface

$$\varepsilon(\uparrow) = \frac{\hbar^2 k^2}{2m} + \mu_B B, \quad \varepsilon(\downarrow) = \frac{\hbar^2 k^2}{2m} - \mu_B B.$$

$$H_e = \frac{p^2}{2m} + g\mu_B \mathbf{B} \cdot \boldsymbol{\sigma}.$$

$$M = -\frac{1}{V} \frac{dE}{dB} = -\frac{\mu_B}{V} (\# \uparrow - \# \downarrow).$$

The relevant area near the Fermi level is

$$\text{Area} \sim \frac{g(\varepsilon_F)}{2} \mu_B B.$$

Thus

$$\# \uparrow \text{ has } \frac{g(\varepsilon_F)}{2} \mu_B B \text{ less,} \quad \# \downarrow \text{ has } \frac{g(\varepsilon_F)}{2} \mu_B B \text{ more.}$$

$$M = \frac{g(\varepsilon_F) \mu_B^2 B}{V}.$$

$$\chi = \frac{\mu_0 \mu_B^2 g(\varepsilon_F)}{V}.$$

Using

$$g(\varepsilon_F) = \frac{3N}{2\varepsilon_F},$$

$$\chi = \mu_0 \mu_B^2 \frac{3N}{2\varepsilon_F}.$$